2015 - 2025: One Decade of Single-Cell

2015
Macosko et al.

2016
10X Chromium
Launches

2017
CITE-seq

2018
Split-seq
Rosenberg et al.

2019
Cell Hashing

2020
Covid-19
Research

2021
Parse launches
WT kits

2022
Genomewide
Perturb-seq

2023
Allen Brain
Atlas

2024
Early
Single Cell
Foundation
Models

2025
Giga-scale Datasets to Power
Drug Discovery

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Mosaic

Measuring how compounds perturb gene function in cells from 100s of patient models in one in vivo experiment at single-cell resolution

Cells from 10s-100s of patient models

Heterogeneous pool of cells

Single organoid / mouse experiment

Single-cell readout

Single-cell RNA seq. + ML & Computational Models

*Cell line, CDX, or PDX**

** Small molecule, biologic, ADC or other modalities*

Drug-induced changes in gene expression in each patient cell

Drug sensitivity for each cell population

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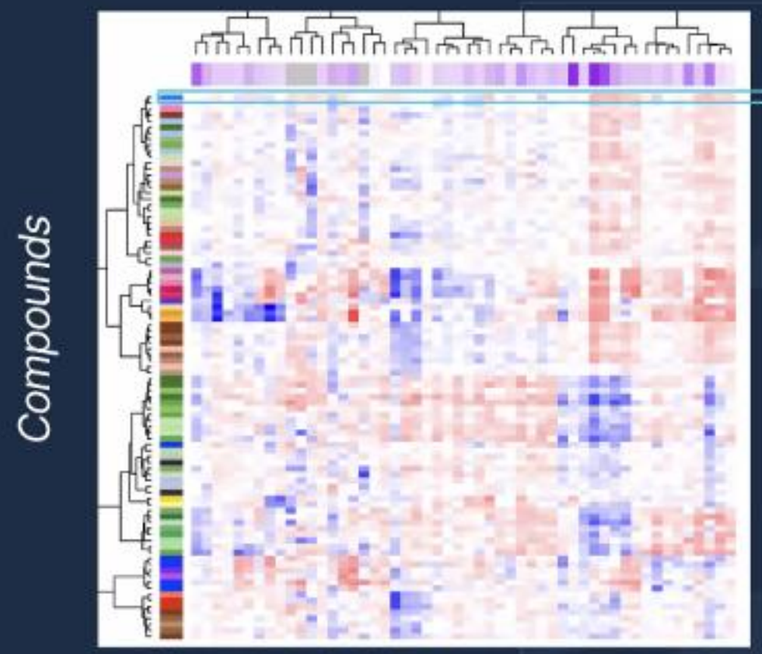
We are using Mosaic to build the largest unbiased, in vivo, single-cell map of how drugs perturb patient biology

We are measuring how 1000s of diverse molecules alter gene function across tumors from 100s of patient models

Unbiased, in vivo, single-cell maps



Each row is a high-content measurement of drug action at scale

1 x	*Drug*
1 x	*Mouse*
50-100 x	*Patient models*
100 x	*Driver genes*
10K x	*Single cells*
20K x	*Gene expressions*

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Chemomes

Gigascale, single-cell maps of how compounds perturb cancer models

BioRxiv

Tahoe-100M

100M

Single cells

50

Cancer models

1.2K

Drug perturbations

50x

Lather than all public data combined

TahoeX1 - A 3 Billion Parameter Virtual Cell Model

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Tahoe-x1: Scaling Perturbation-Trained Single-Cell Foundation Models to 3 Billion Parameters

Shreshth Gandhi[*,†,‡,1], Farnoosh Javadi[*,1], Valentine Svensson[*,1], Umair Khan[*,1], Matthew G. Jones[1], John Yu[†,1], Daniele Merico[†,1], Hani Goodarzi[†,1], Nima Alidoust[†,1]

[1]Tahoe Therapeutics.

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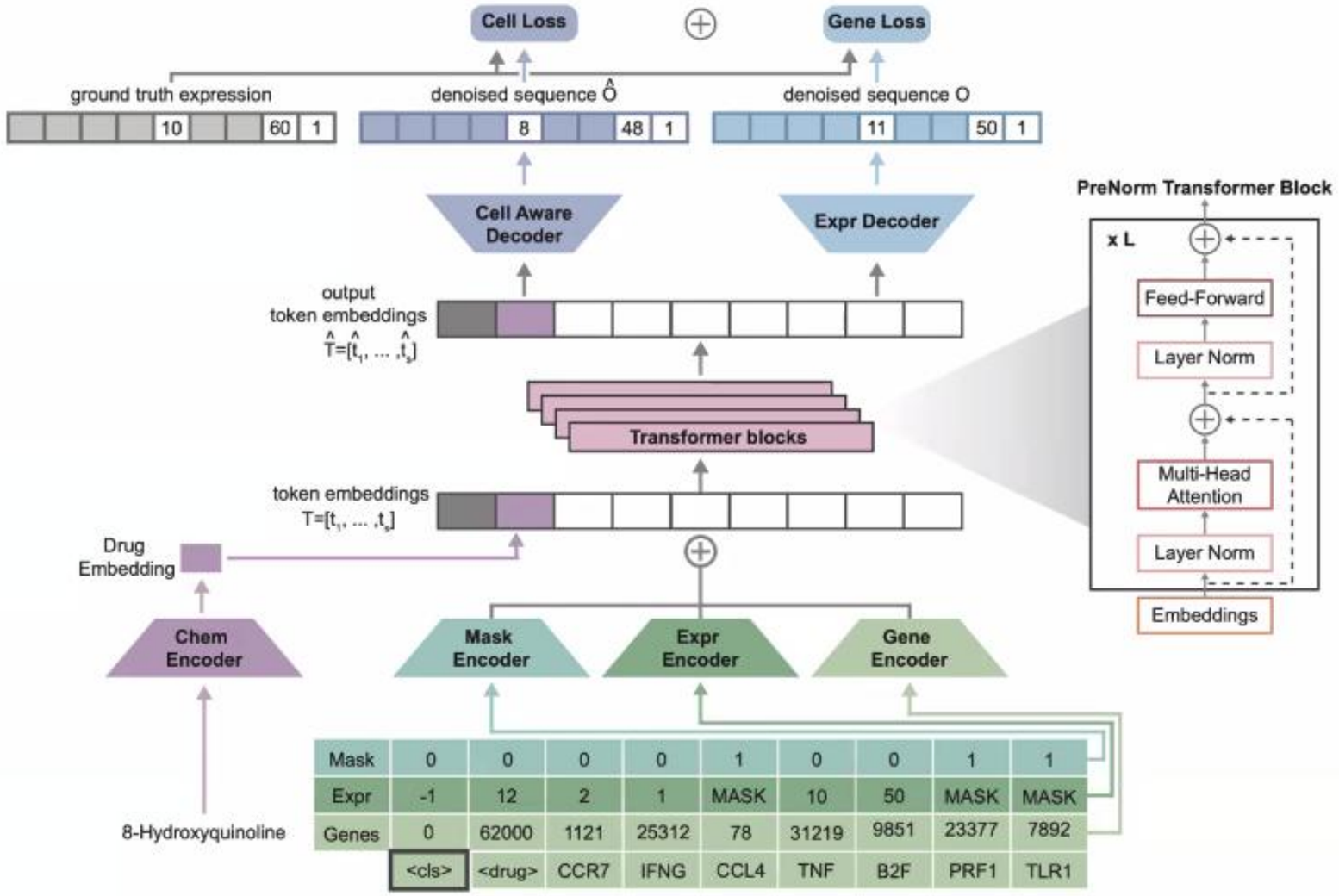
The Tahoe-x1 model

Two self-supervised objectives: Cell-loss trains the **cell-embedding** and gene-loss trains the **contextual gene-embeddings.**

Applications of the model use either cell-level embeddings or contextual gene-embeddings.



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TahoeX1: Lower Training Costs leads to Better Models

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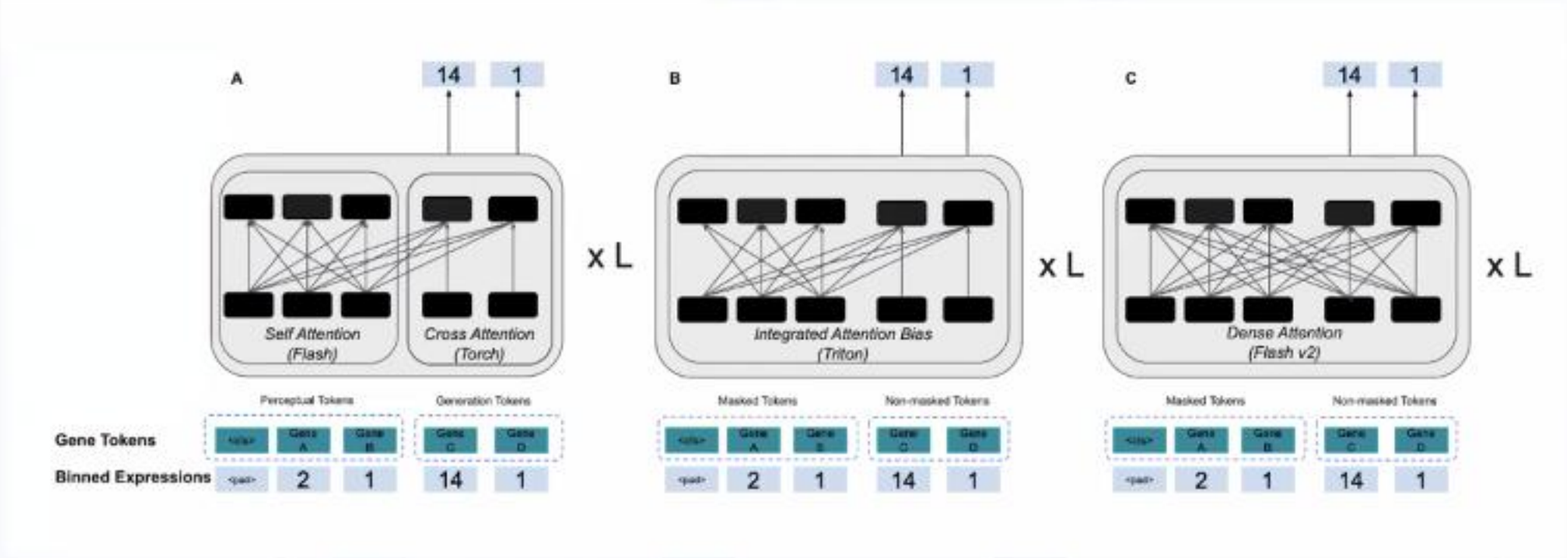
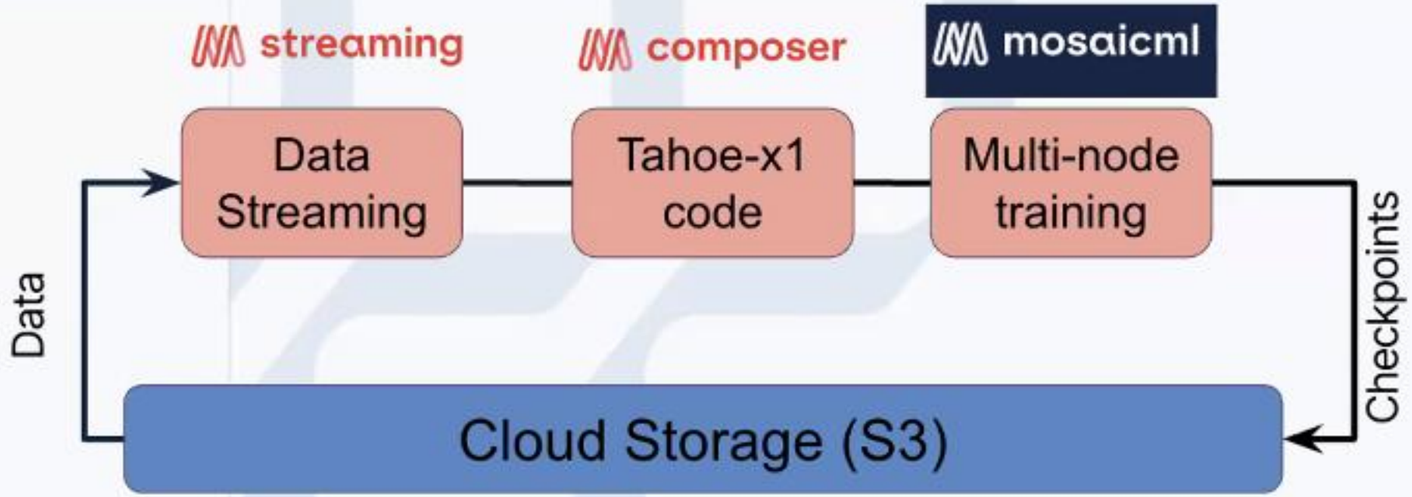
Tahoe-x1: Key ingredients for efficiency

- MosaicML Streaming data-loader + FSDP → **Eliminates data-loader bottlenecks in training.**
- Flash-attention-v2, bf16 mixed precision training, low-precision layernorm → **Ensures high GPU utilization**





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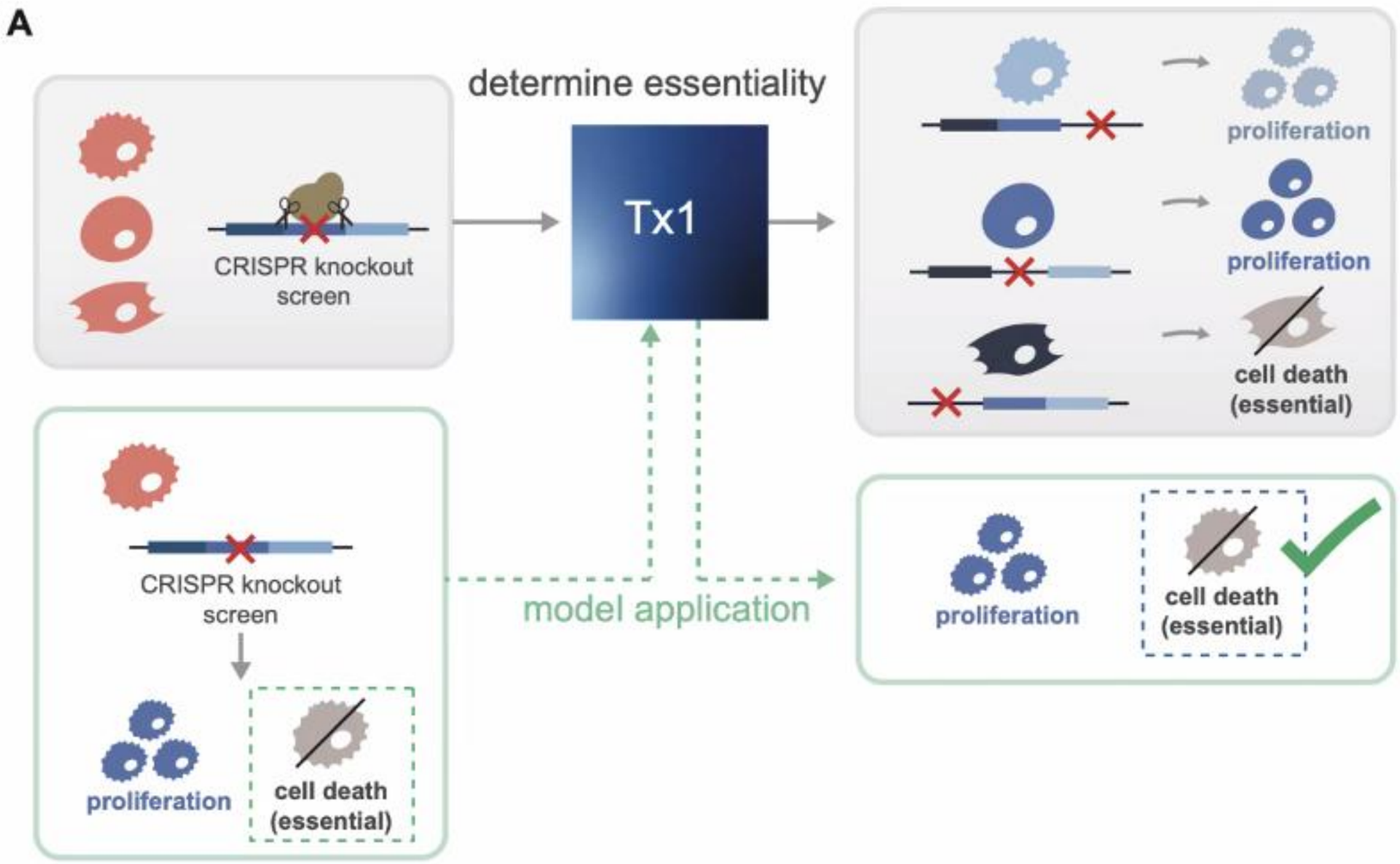
Application 1: Predicting Gene-Essentiality on DepMap

Task: Given only the gene-expression of a cell line **predict which genes are essential for cell survival in that specific cell-line.**

Target: Gene dependency scores derived from CRISPR gene knockout screens.

Input: Contextual gene-embeddings using CCLE bulk expression data for the corresponding Depmap cell-lines.

TAHOE



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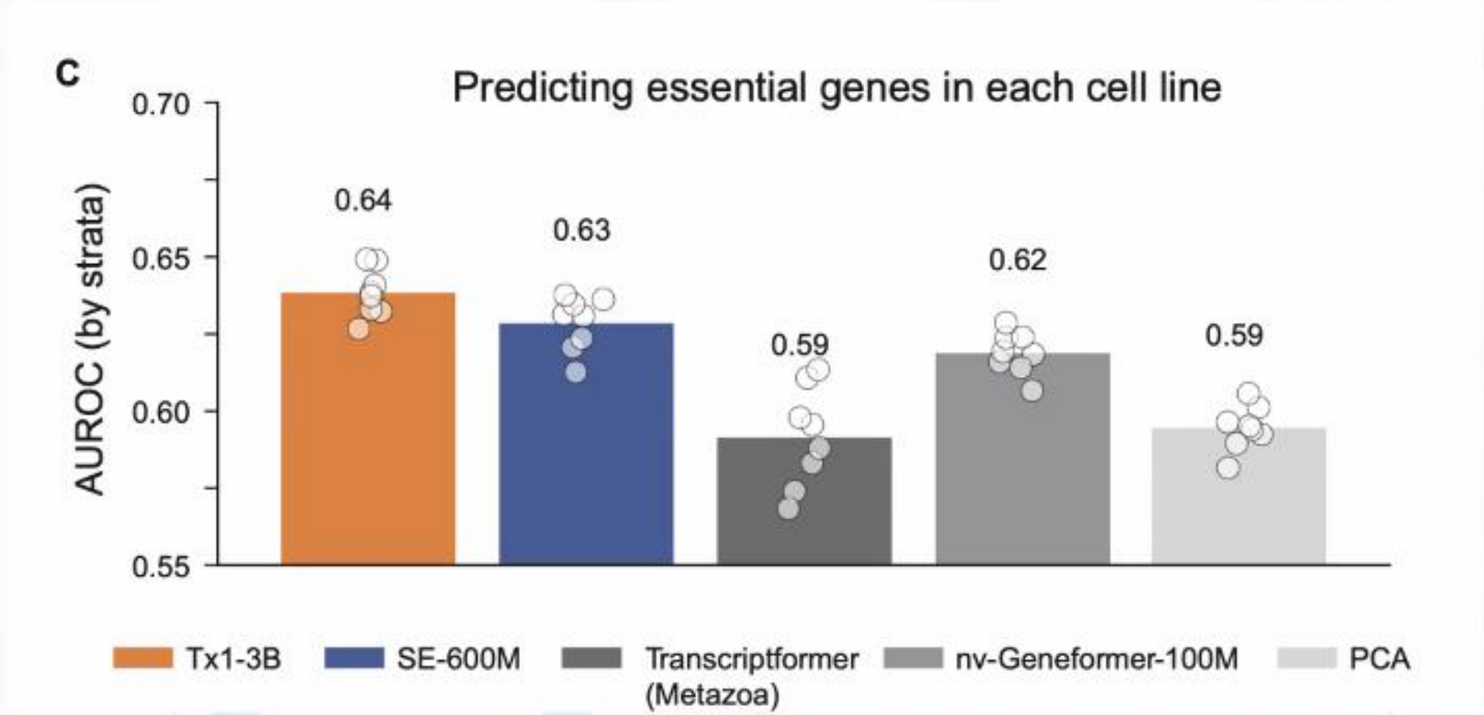

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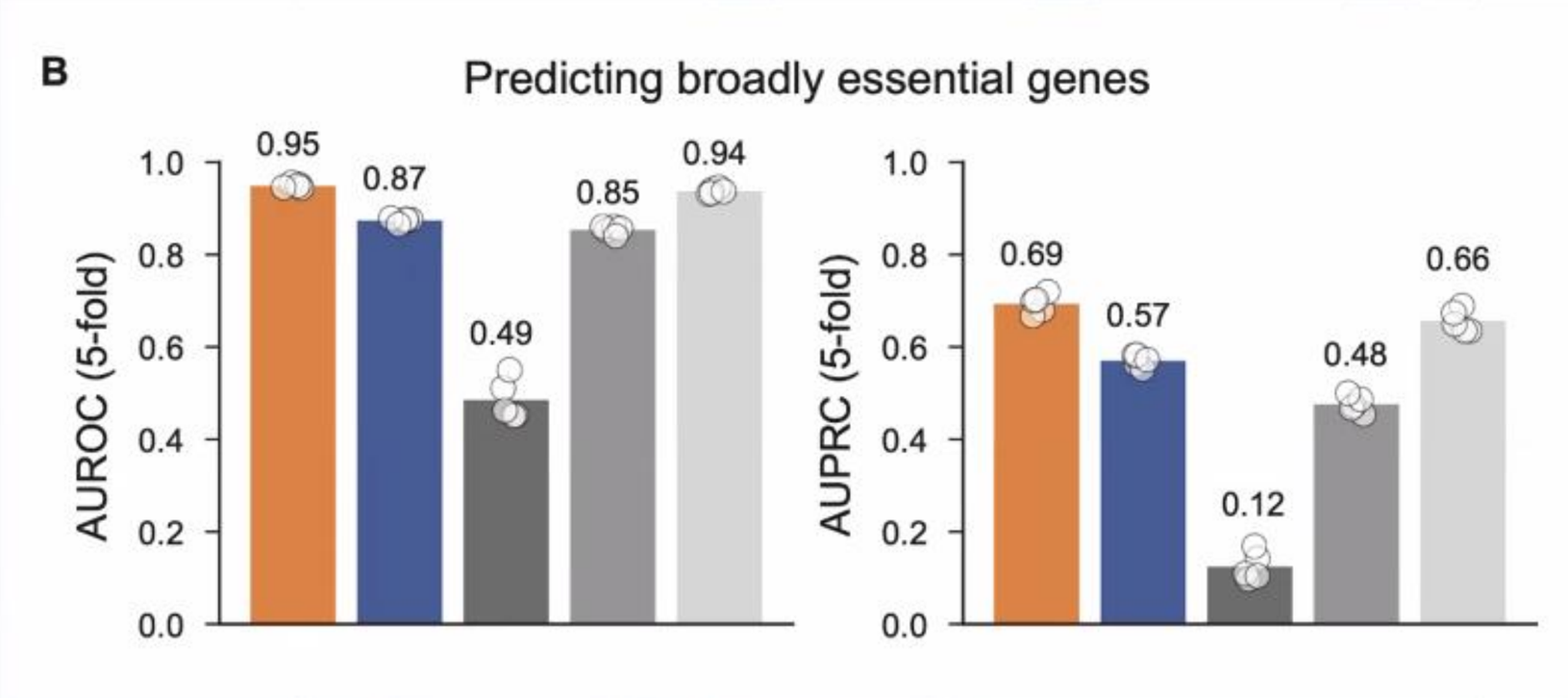
Application 1: Predicting Gene-Essentiality on DepMap

Task: Given the gene-expression of all cell-lines predict **which genes are essential in a majority of cell-lines**

Target: Gene dependency scores aggregated across cell-lines

Input: Contextual gene-embeddings averaged over cell-lines



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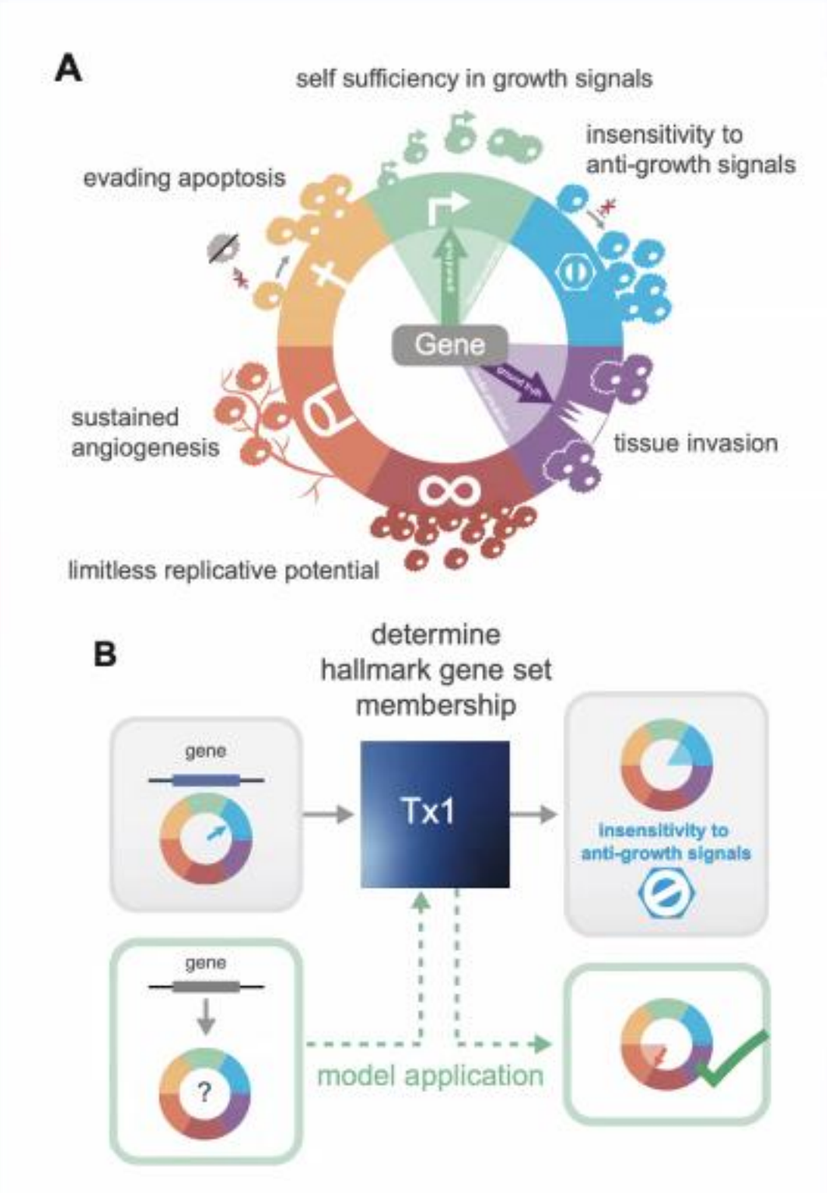
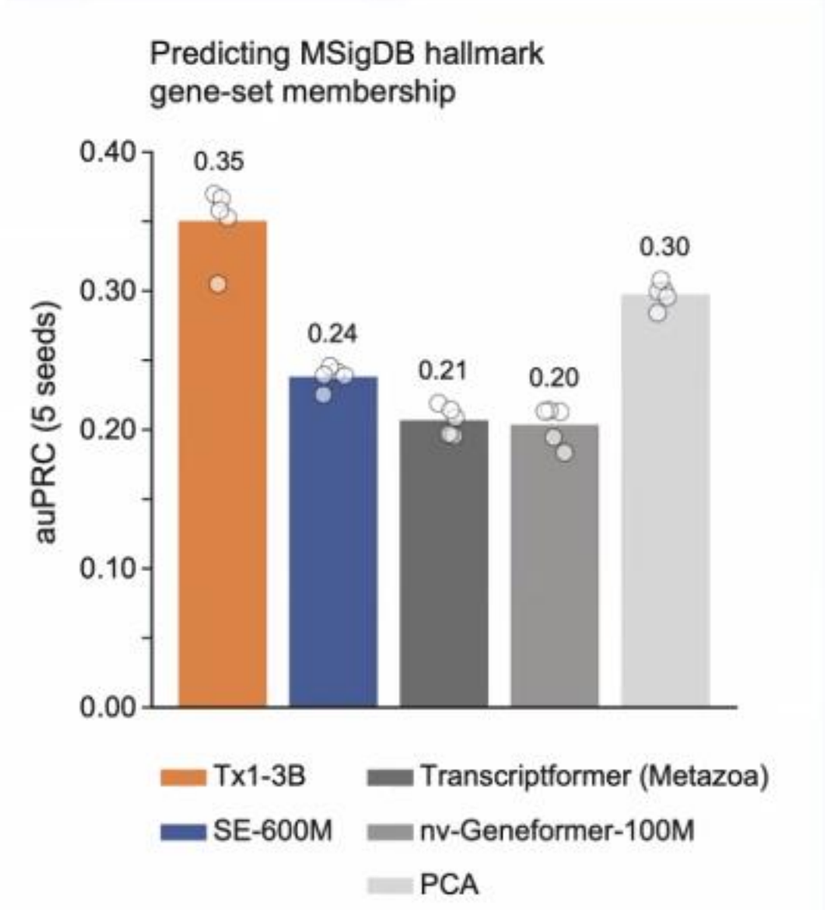
Application 2: Predicting MSigDB hallmark membership

Task: Given a gene, predict the functional categories to which it belongs.

Target: Gene-set membership labels derived from cancer hallmarks

Input: Contextual gene-embeddings averaged over a panel of cell-lines





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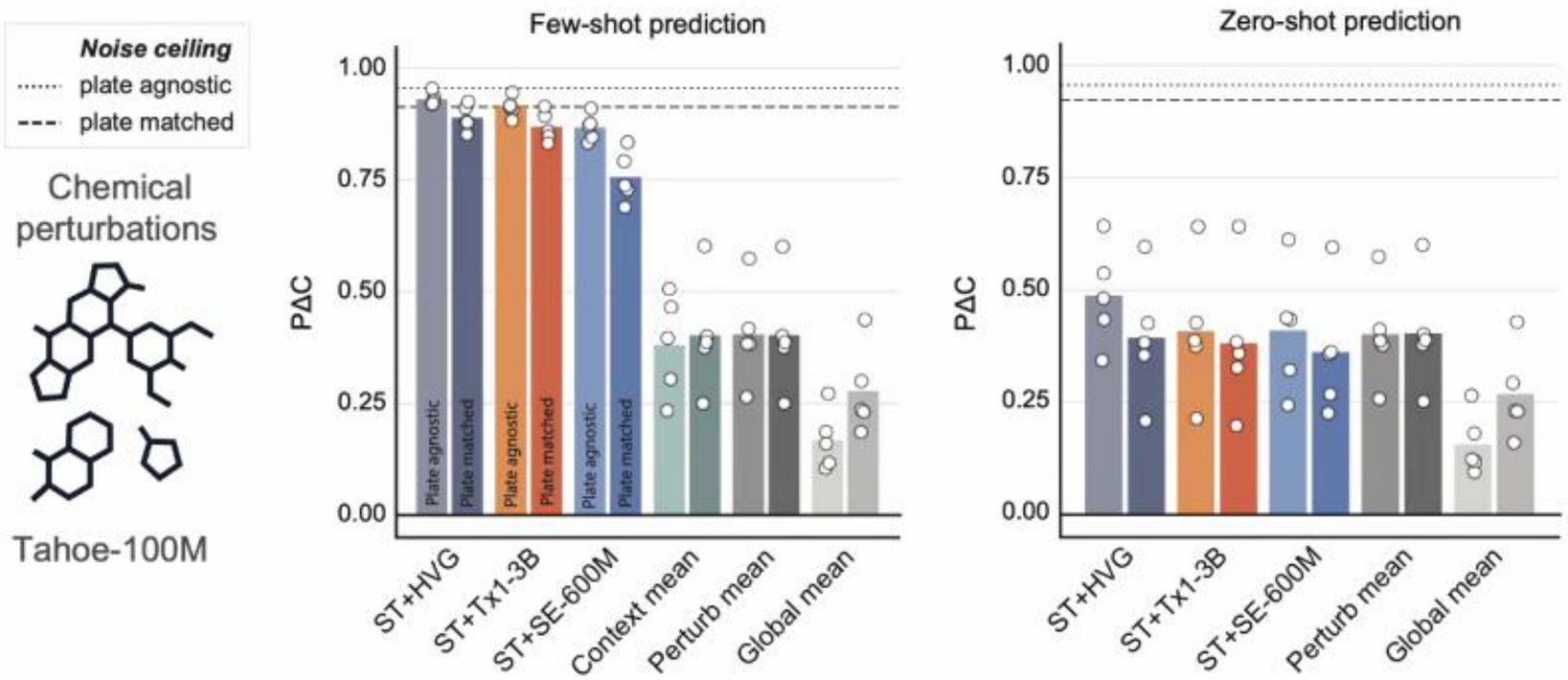
Application 3: Predicting the effect of perturbations in new contexts

Task: Predict the effect of a given perturbation in a new context

Target: Post-perturbation expression for 2000 HVGs

Input: Cell-embeddings for all cells in the dataset.



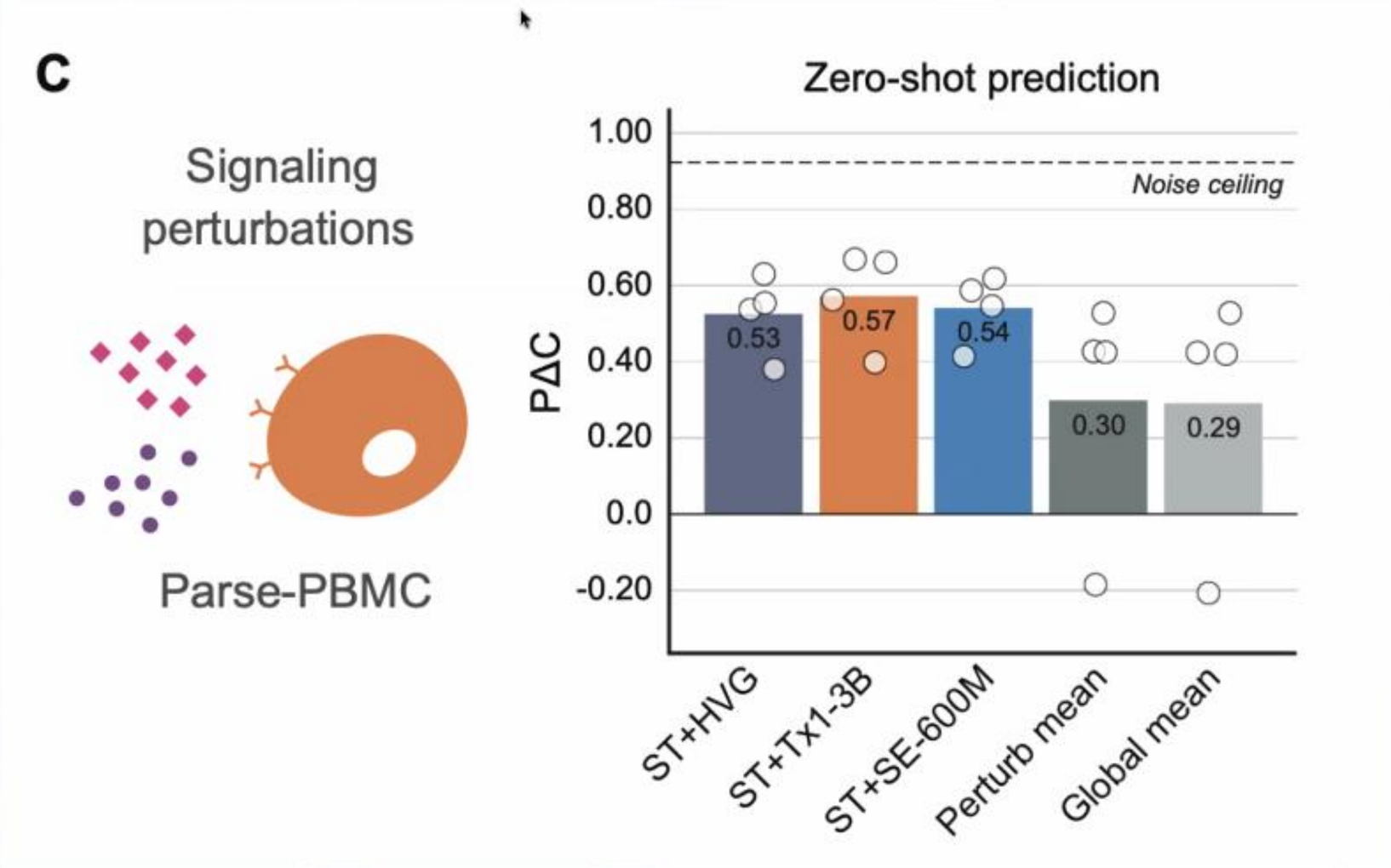
TAHOE

Application 3: Predicting the effect of perturbations in new contexts

Task: Predict the effect of a given perturbation in a new context **on a new dataset**

Target: Post-perturbation expression for 2000 HVGs

Input: Cell-embeddings for all cells in the dataset+ST model pre-trained on Tahoe-100M

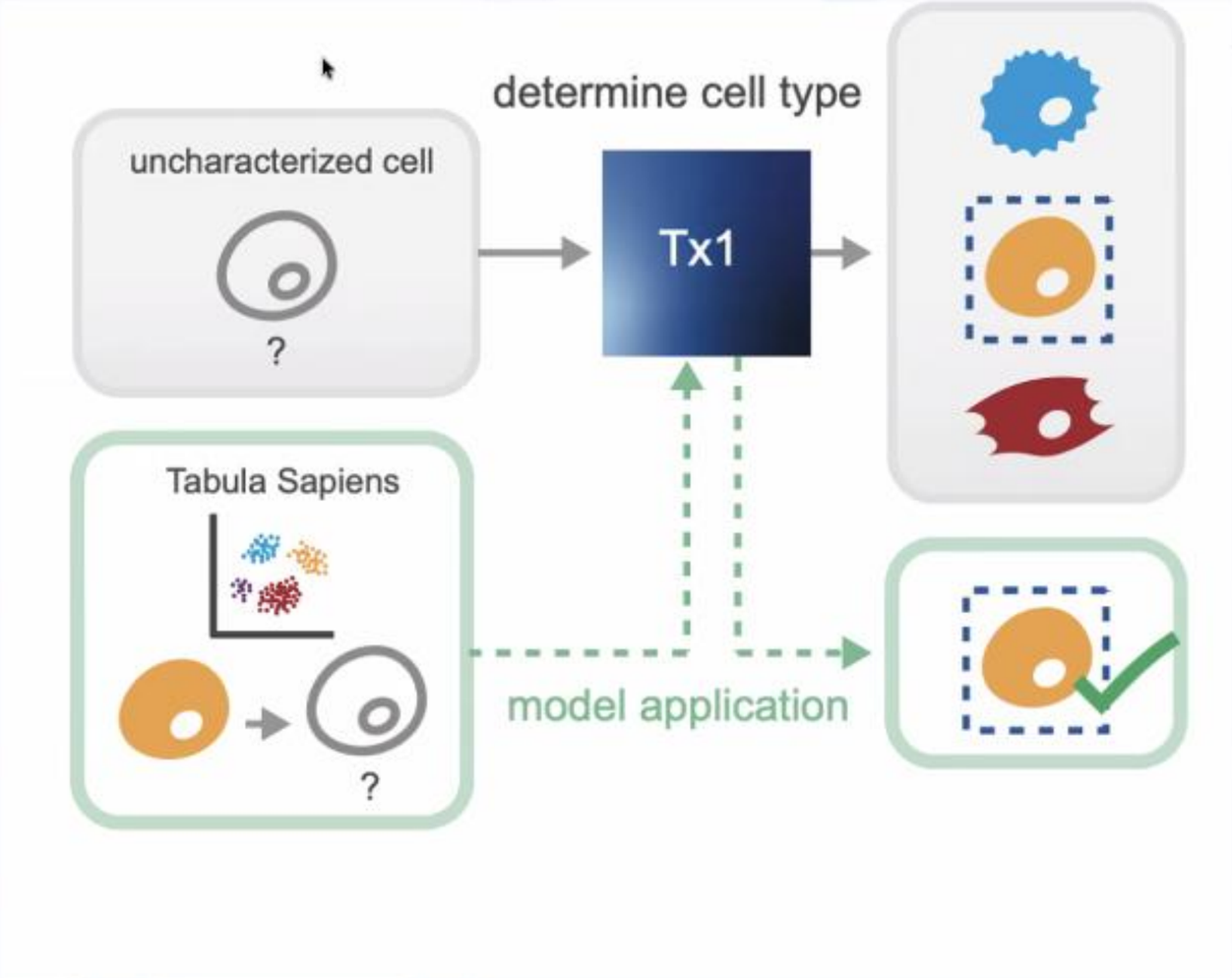


TAHOE

Application 4: Classifying cell-types

Task: Assign cell-type identities based on single-cell gene expression profiles.

Input: Cell-embeddings for individual cells



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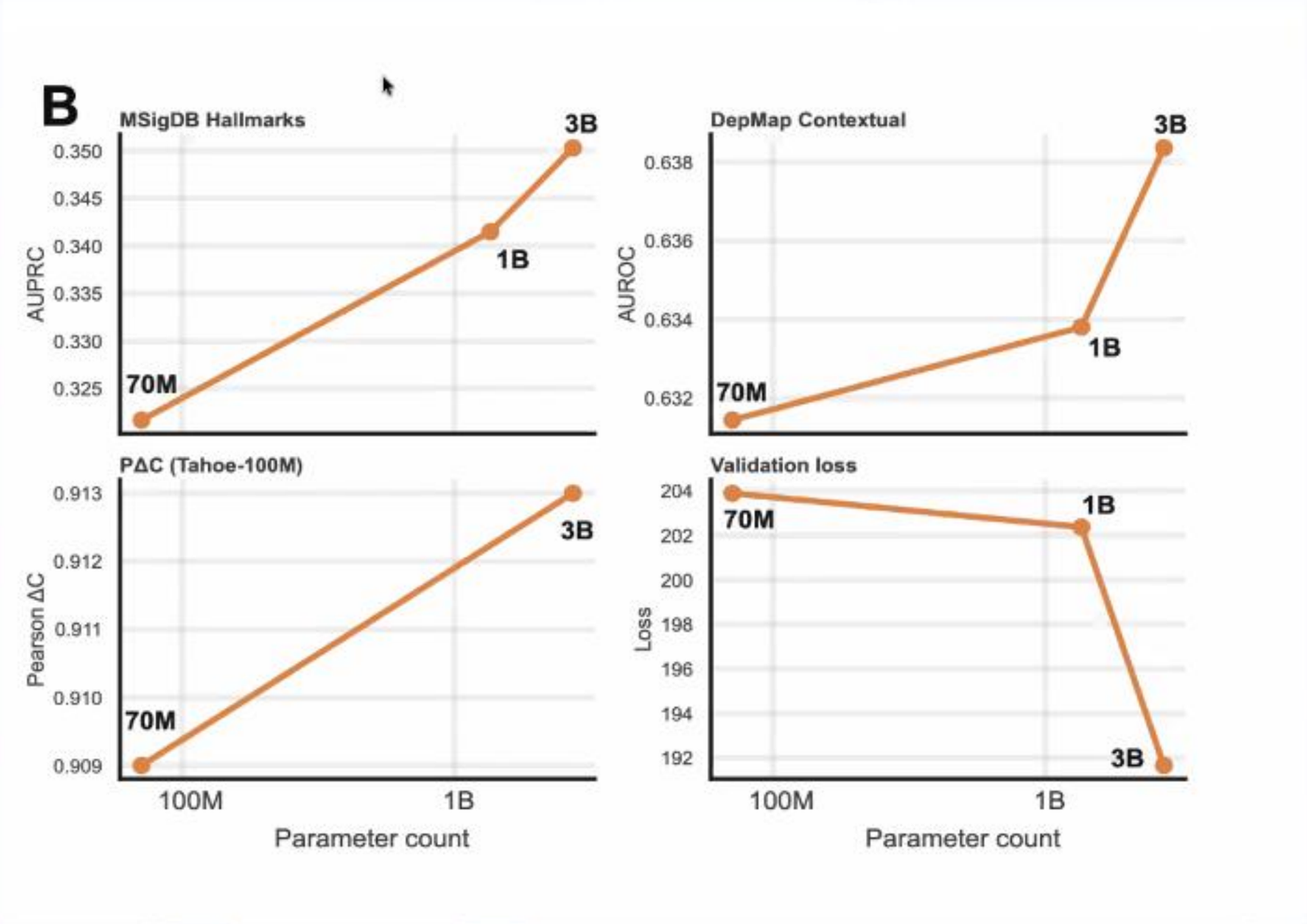
Scaling behaviour

- We consistently saw improved performance on downstream benchmarks when scaling up the size of the embedding model.



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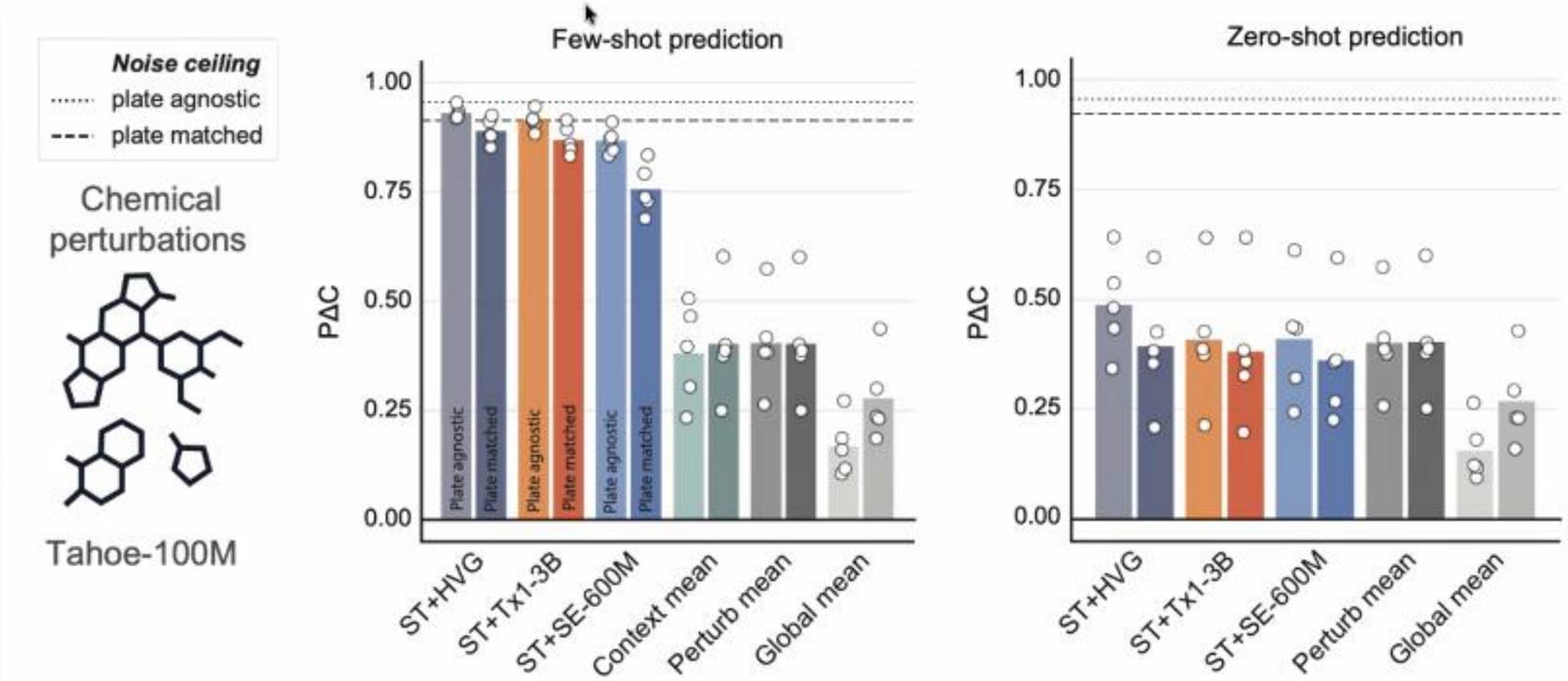
Key gap: Zero shot performance close to baselines

Task: Predict the effect of a given perturbation in a new context

Target: Post-perturbation expression for 2000 HVGs

Input: Cell-embeddings for all cells in the dataset.



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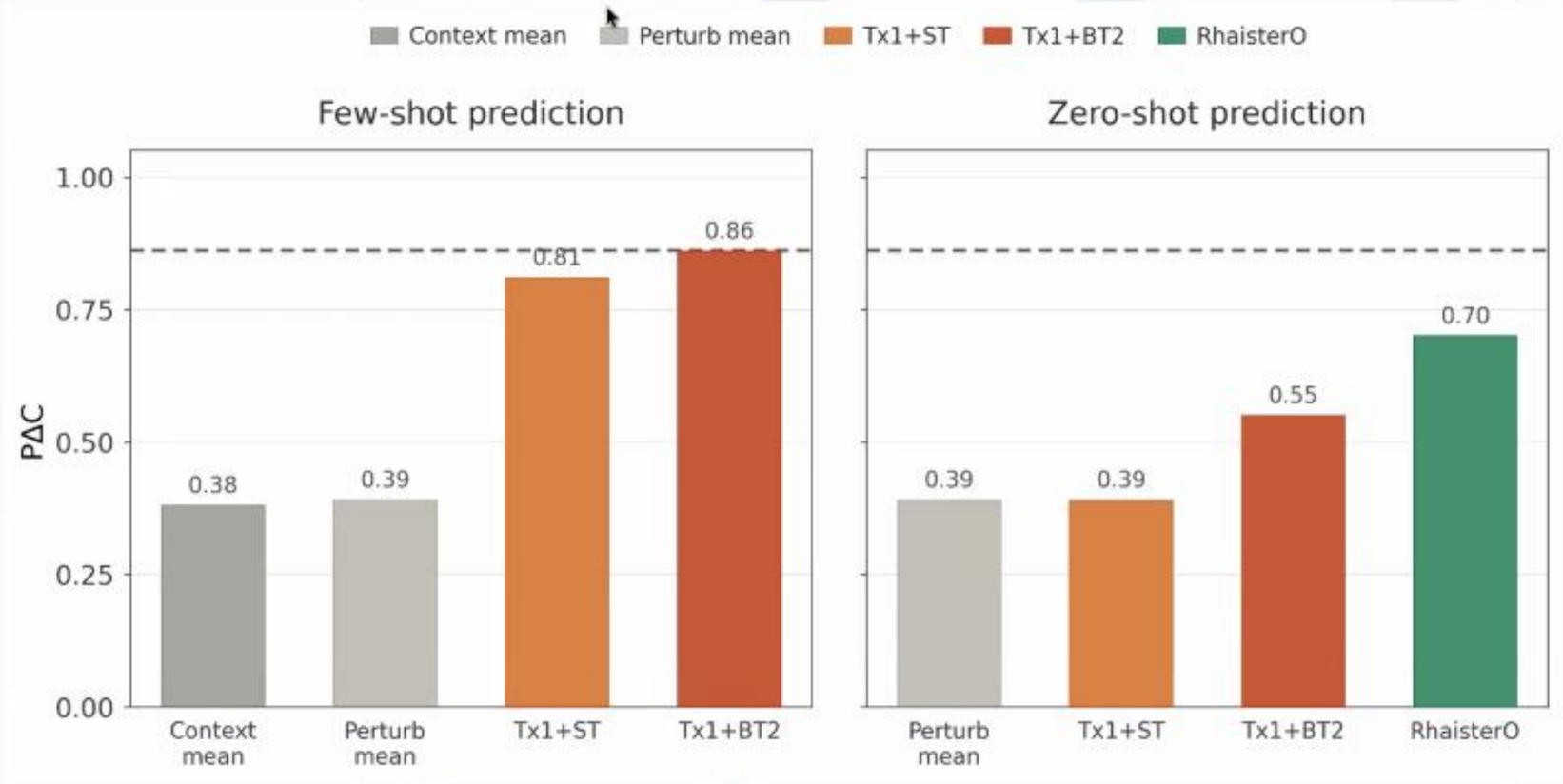
Current generation Tx1 powered models greatly improve on perturbation preduction

Task: Predict the effect of a given perturbation in a new context

Target: Post-perturbation expression for 2000 HVGs

Input: Cell-embeddings for all cells in the dataset.



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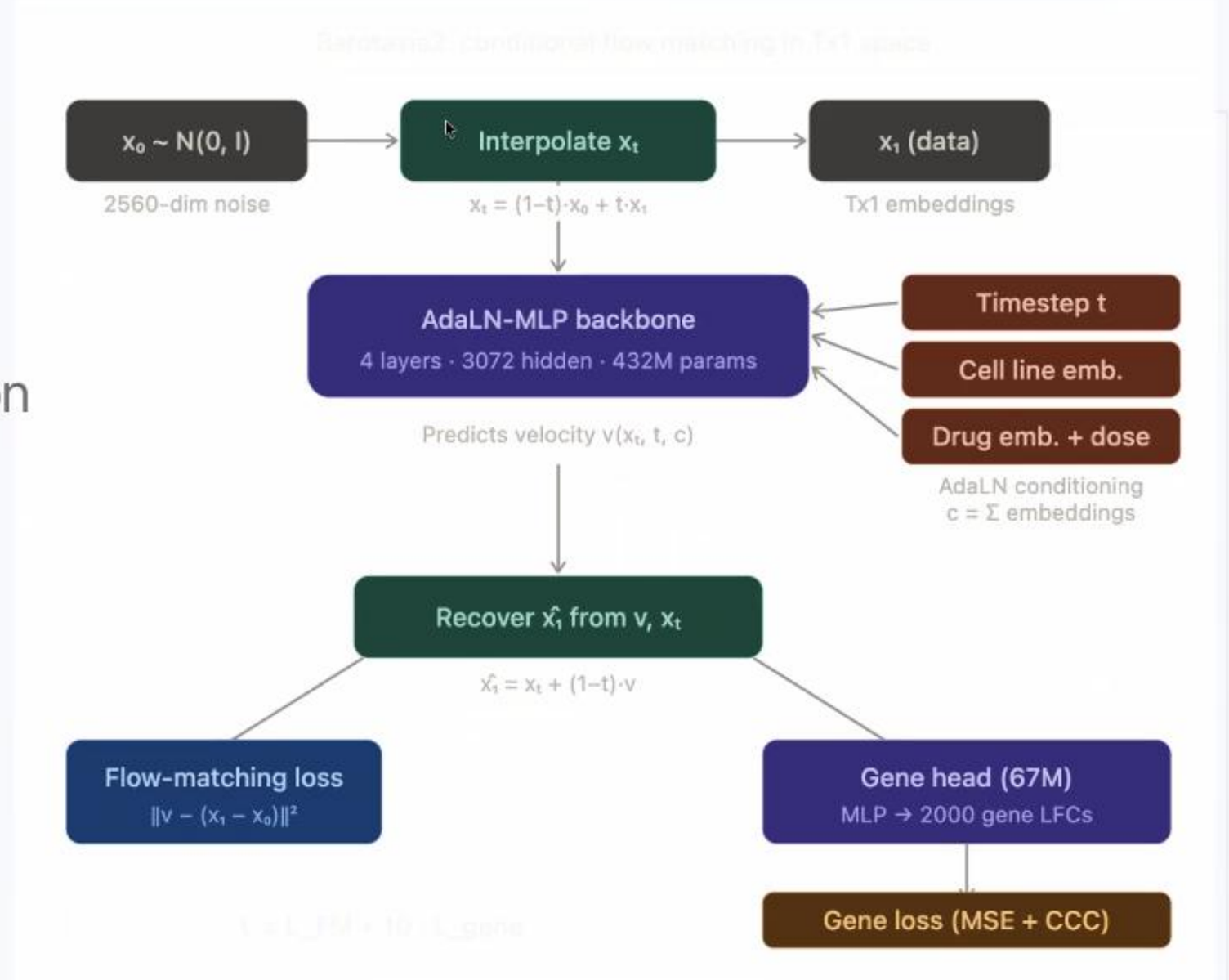
BT2: Conditional generative model in Tx1 embedding space

- Does not require pairs of cell sets (control-perturbed).
- 25x faster to train than the ST model on Tahoe-100M data (~3hrs/experiment)



Rhaister: Auto-Research on Tx1 embeddings ~5mins/experiment

Andrej Karpathy
@karpathy

I packaged up the "autoresearch" project into a new self-contained minimal repo if people would like to play over the weekend. It's basically nanochat LLM training core stripped down to a single-GPU, one file version of ~630 lines of code, then:

- the human iterates on the prompt (.md)
- the AI agent iterates on the training code (.py)

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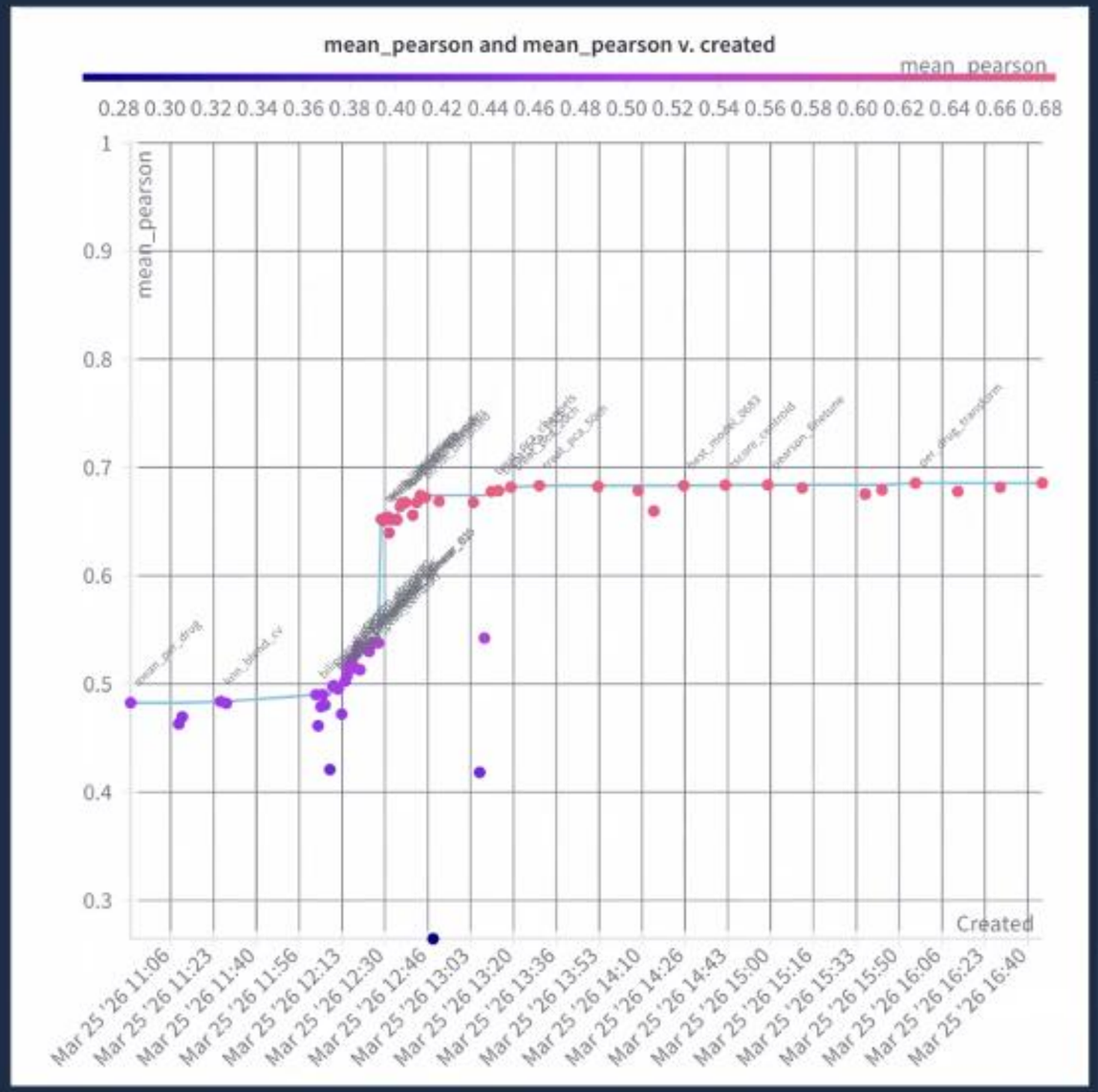
Rhaister: Auto-Research on Tx1 embeddings ~5mins/experiment



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Open questions and future directions

- Generalization to entirely new experimental conditions - reducing domain drift
- Combining LLM reasoning models with Tx1 → interpretability
- Experimental validation → beyond immortalized cell lines
- Sandboxed agentic exploration of architectures → exponential self-improvement

Try it yourself - no installation required!

https://tinyurl.com/tx1-berkeley

Tx1-70M Embedding Demo

This notebook demonstrates how to use the **Tx1-70M** model via the public Gradio space to generate embeddings from your AnnData objects.

Gradio Space: https://huggingface.co/spaces/tahoebio/tx1-demo

What you'll learn:

- Upload your AnnData file to the Gradio space
- Get back 1024-dimensional embeddings
- Download results as parquet or h5ad
- Visualize UMAP projections

Requirements:

- AnnData file (.h5ad) with gene expression data
- Gene IDs as Ensembl IDs (ENSG*) or gene symbols
- Max 100,000 cells (will auto-subsample if larger)
- Max file size: 5GB

Models 3

tahoebio/Tahoe-x1

3B • Updated Oct 24, 2025 • 23 • 67

tahoebio/Tahoe-x1-embeddings

Viewer • Updated Dec 3, 2025 • 95.6M • 430 • 3

Tx1 models (3B,1B,70M) are all fully open source + open weights